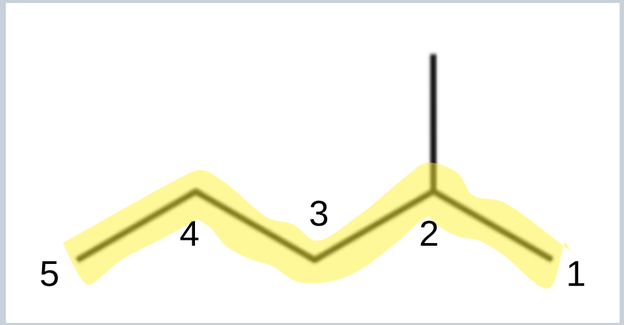
Card A01 - Foundation: C_6H_{14}

Numbered Parent

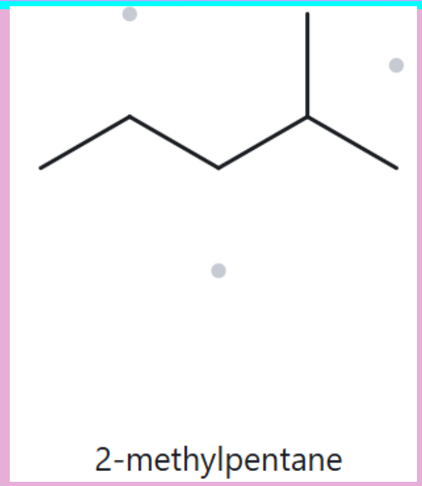


Naming Trail

Parent: Longest chain = 5 C → pentane, chain.
Numbering: Left to right gives 4; right to left gives 2. First point of difference: 2 < 4, so number from the right.
Substituent: Methyl on C2 → 2-methyl

My Name

2-methylpentane



The Alkane Check

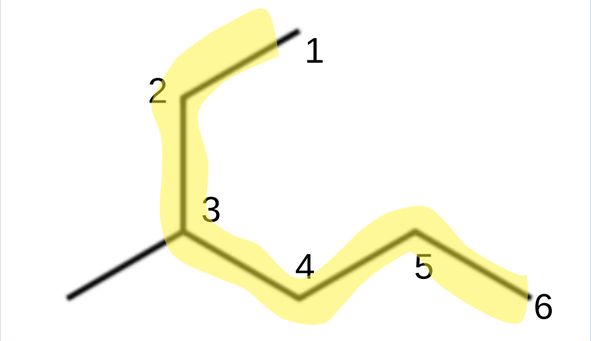
Returned name:
2-methylpentane
✓ Match

Verification Strip

✓longest chain confirmed ✓lowest locant ✓alphabetical order ✓punctuation ✓formula counted

Card A02 - Foundation: C_7H_{16}

Numbered Parent

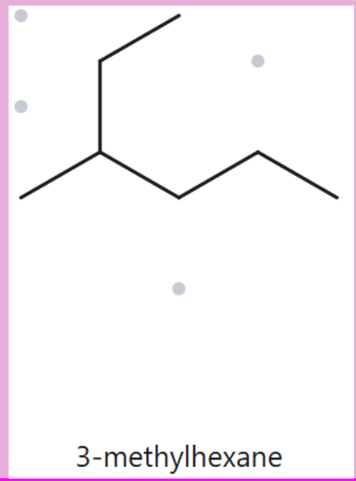


Naming Trail

Parent: Longest chain = 6 C → hexane, chain.
Numbering: Top end to bottom-right gives 3; bottom-right end to top gives 4. First point of difference: 3 < 4, so number from the top end.
Substituent: Methyl on C3 → 3-methyl

My Name

3-methylhexane



The Alkane Check

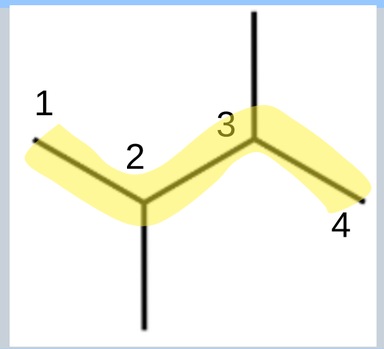
Returned name:
3-methylhexane
✓ Match

Verification Strip

✓longest chain confirmed ✓lowest locant ✓alphabetical order ✓punctuation ✓formula counted

Card A03 - Foundation: C_6H_{14}

Numbered Parent

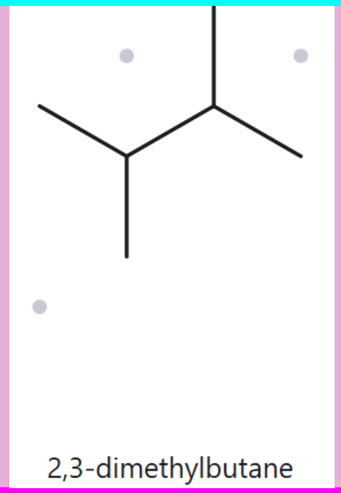


Naming Trail

Parent: Longest chain = 4 C → butane, chain. Two C1 and two C4 paths exist, but each has two methyl groups, so they give the same name.
Numbering: Left to right gives 2,3; right to left also gives 2,3. Both directions give identical locants (2,3), so either direction works.
Substituent: Methyl on C2 and C3 → 2,3-dimethyl

My Name

2,3-dimethylbutane



The Alkane Check

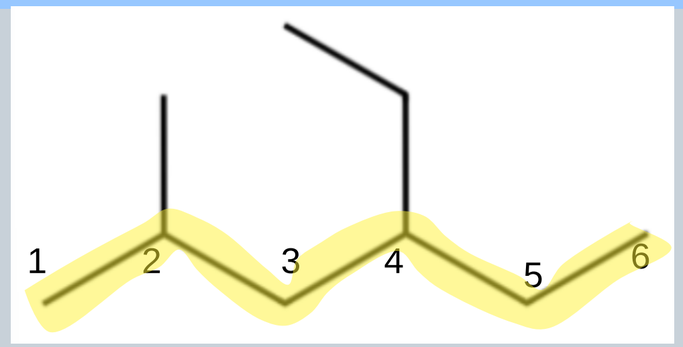
Returned name:
2,3-dimethylbutane
✓ Match

Verification Strip

✓longest chain confirmed ✓lowest locant ✓alphabetical order ✓punctuation ✓formula counted

Card A04 - Developing: C_9H_{20}

Numbered Parent

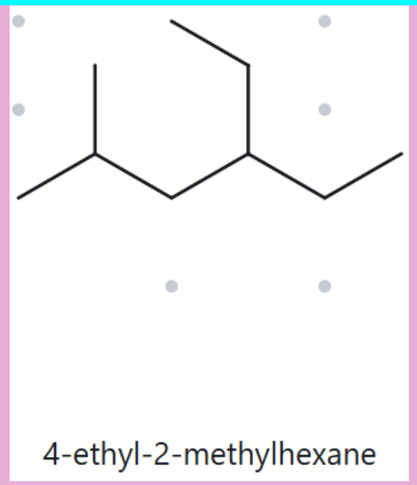


Naming Trail

Parent: Longest chain = 6 C → hexane, chain.
Numbering: Left to right gives 2,4; right to left gives 3,5.
First point of difference: 2 < 3, so number from left to right.
Substituents: Ethyl on C4, methyl on C2 → ordered alphabetically as 4-ethyl-2-methyl.

My Name

4-ethyl-2-methylhexane



The Alkane Check

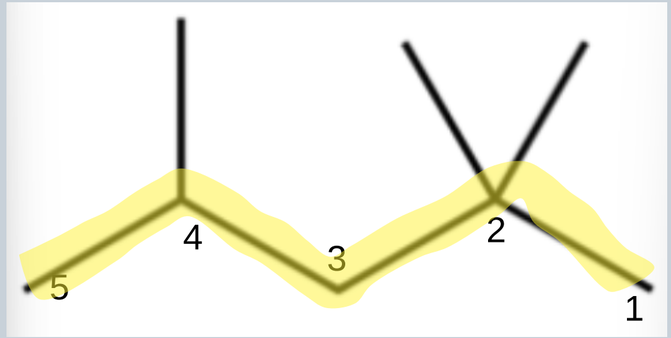
Returned name:
4-ethyl-2-methylhexane
✓ Match

Verification Strip

✓longest chain confirmed ✓lowest locant ✓alphabetical order ✓punctuation ✓formula counted

Card A05 - Developing: C_8H_{18}

Numbered Parent

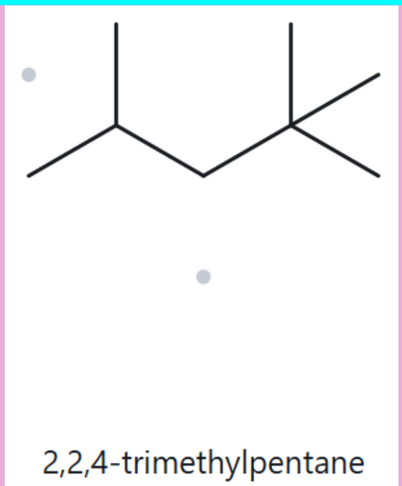


Naming Trail

Parent: Longest chain = 5 C → pentane, chain.
Numbering: Left to right gives 2,4,4; right to left gives 2,2,4. First point of difference: 2 < 4, so number from right to left.
Substituents: Methyl on C2, C2, and C4 → 2,2,4-trimethyl.

My Name

2,2,4-trimethylpentane



The Alkane Check

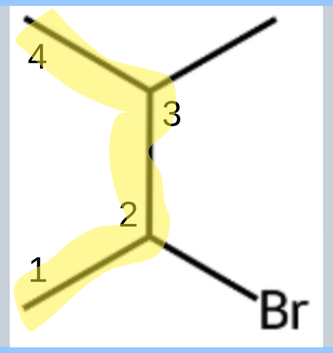
Returned name:
2,2,4-trimethylpentane
✓ Match

Verification Strip

✓longest chain confirmed ✓lowest locant ✓alphabetical order ✓punctuation ✓formula counted

Card A06 - Developing: $C_5H_{11}Br$

Numbered Parent



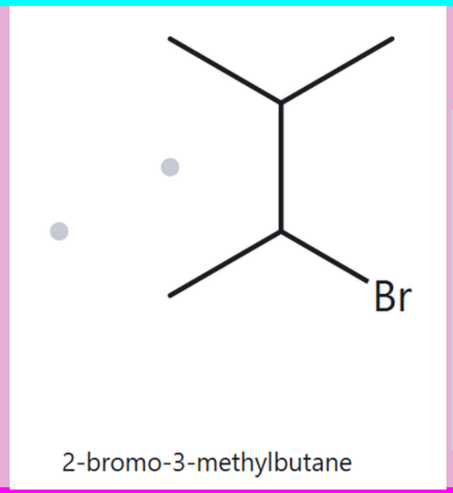
Naming Trail

Parent: Longest chain = 4 C → butane, chain.
Numbering: Bottom-left to top gives 2,3; top to bottom-left also gives 2,3. The lower number (2) goes to bromo due to alphabetical priority ('b' < 'm').
Substituents: Bromo on C2, methyl on C3 → ordered alphabetically as 2-bromo-3-methyl.

My Name

2-bromo-3-methylbutane

The Alkane Check



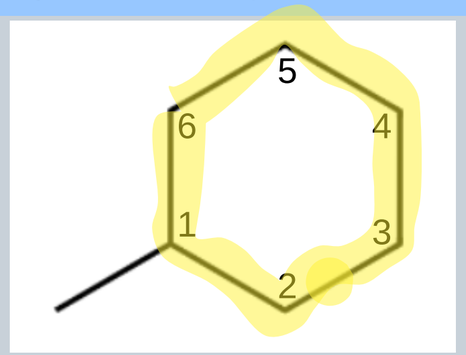
Returned name:
2-bromo-3-methylbutane
✓ Match

Verification Strip

✓longest chain confirmed ✓lowest locant ✓alphabetical order ✓punctuation ✓formula counted

Card A01 - Developing: C_7H_{14}

Numbered Parent

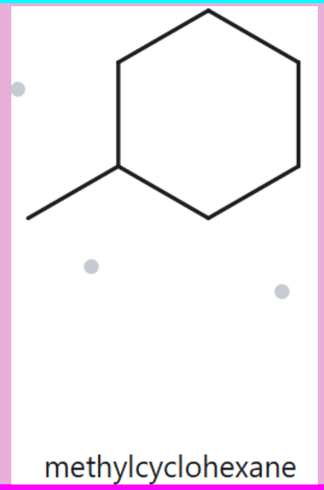


Naming Trail

Parent: 6-carbon ring → cyclohexane, ring.
Numbering: The substituted carbon is automatically C1. Because it is the only substituent on the ring, the locant number "1" is omitted from the final name.
Substituents: Methyl on ring → methyl.

My Name

methylcyclohexane



The Alkane Check

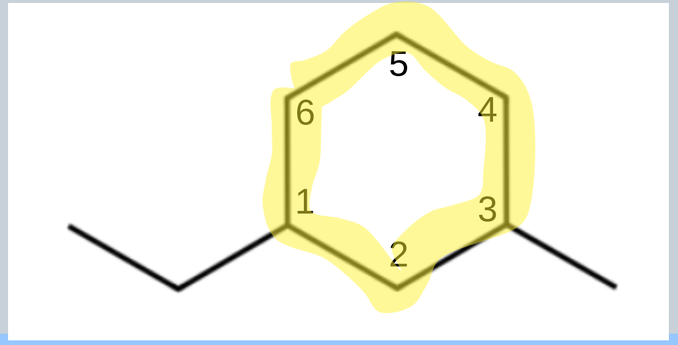
Returned name:
methylcyclohexane
✓ Match

Verification Strip

✓longest chain confirmed ✓lowest locant ✓alphabetical order ✓punctuation ✓formula counted

Card A08 - Challenge: C_9H_{18}

Numbered Parent

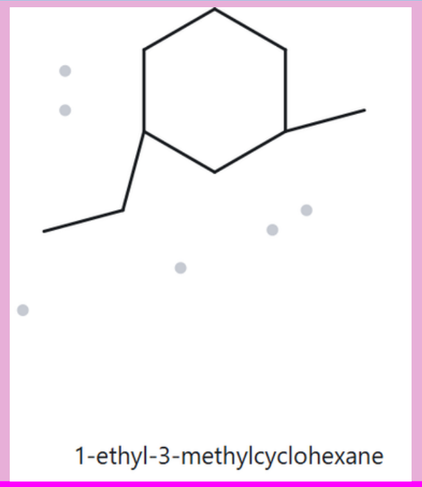


Naming Trail

Parent: 6-carbon ring → cyclohexane, ring.
Numbering: Counting through the bottom vertex gives locants 1,3 regardless of which substituent you start from. Since both directions yield the identical locant set (1,3), 1 is assigned to ethyl due to alphabetical priority ('e' < 'm').
Substituents: Ethyl on C1, methyl on C3 → ordered alphabetically as 1-ethyl-3-methyl.

My Name

1-ethyl-3-methylcyclohexane



The Alkane Check

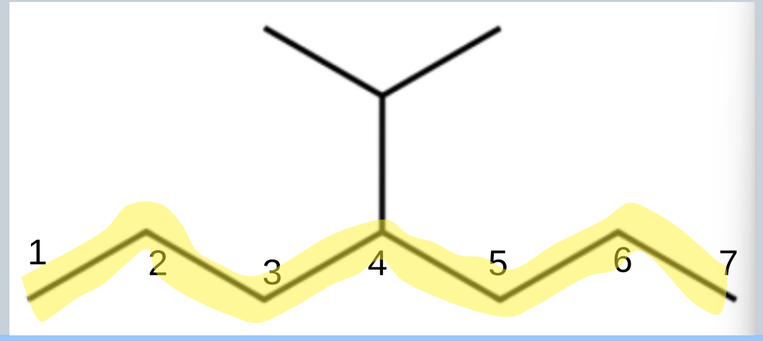
Returned name:
1-ethyl-3-methylcyclohexane
✓ Match

Verification Strip

✓longest chain confirmed ✓lowest locant ✓alphabetical order ✓punctuation ✓formula counted

Card A09 - Challenge: $C_{10}H_{22}$

Numbered Parent

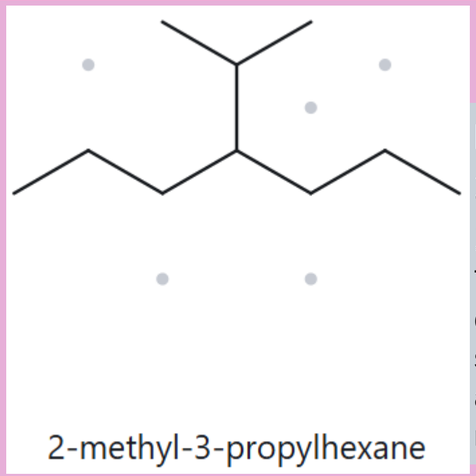


Naming Trail

Parent: Longest chain = 7 C → heptane, chain.
Numbering: Left to right gives 4; right to left also gives 4. Both directions yield identical locants (4), so either direction works (shown left to right).
Substituents: Isopropyl on C4 → 4-isopropyl.

My Name

4-isopropylheptane



The Alkane Check

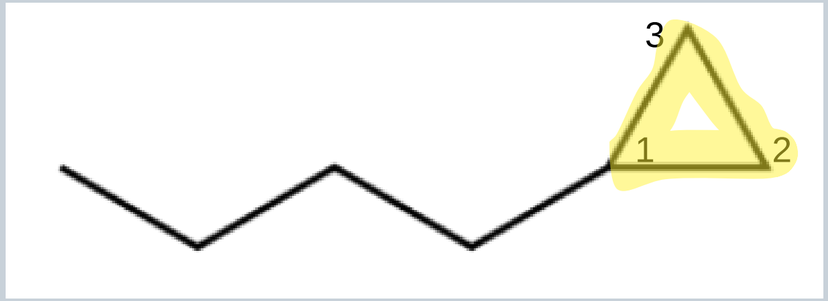
Returned name:
2-methyl-3-propylhexane
✘ Not match
The Alkane Tool incorrectly chose a 6-carbon chain (hexane) to maximize substituent count. Chain length takes absolute priority over substituent count, making the 7-carbon heptane chain correct.

Verification Strip

✓longest chain confirmed ✓lowest locant ✓alphabetical order ✓punctuation ✓formula counted

Card A10 - Challenge: C_7H_{14}

Numbered Parent



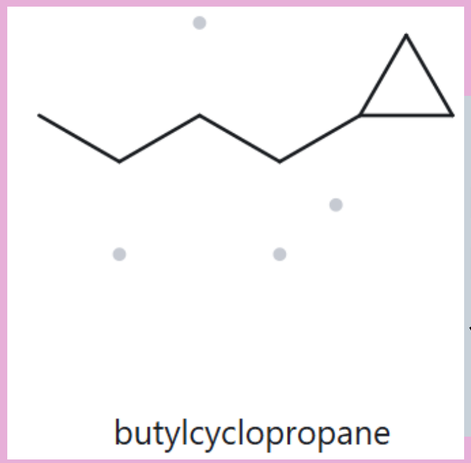
Naming Trail

Parent: Open chain has 4 C vs. ring's 3 C → cyclopropane, ring (activity specifies that the ring will be the parent).
Numbering: The ring carbon bearing the butyl group is C1. Both directions around the ring give the same locant, and with only one substituent the locant “1” is omitted.
Substituents: Butyl on C1 → butyl.

My Name

butylcyclopropane

The Alkane Check



Returned name:
butylcyclopropane
✓ Match

Verification Strip

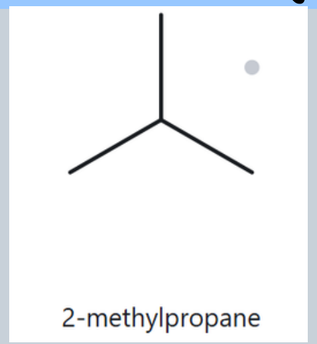
✓longest chain confirmed ✓lowest locant ✓alphabetical order ✓punctuation ✓formula counted

Card B01 - Foundation: 2-methylpropane

Name Decoded

Parent: Propane → a 3-carbon chain.
Substituents: methyl on C2
Expected Formula: 3+1=4 C; open-chain alkane
→ C_nH_{2n+2} → **C_4H_{10}**

Bond-line Structure (The Alkane)



Formula Check

Counted from the drawing: C_4H_{10}
Acyclic alkane: C_nH_{2n+2} = C_4H_{10} ✓

Back-Check

Returned by The Alkane: 2-methylpropane
✓ Same as the given name.

Verification Strip

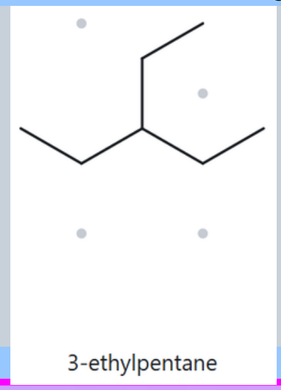
✓parent length matches ✓substituent positions match ✓formula agrees ✓back-check

Card B02 - Foundation: 3-ethylpentane

Name Decoded

Parent: Pentane → a 5-carbon chain.
Substituents: ethyl on C3.
Expected Formula: 5+2=7 C; open-chain alkane → C_nH_{2n+2} → **C_7H_{16}**

Bond-line Structure (The Alkane)



Formula Check

Counted from the drawing: C_7H_{16}
Acyclic alkane: C_nH_{2n+2} = C_7H_{16} ✓

Back-Check

Returned by The Alkane: 3-ethylpentane
✓ Same as the given name.

Verification Strip

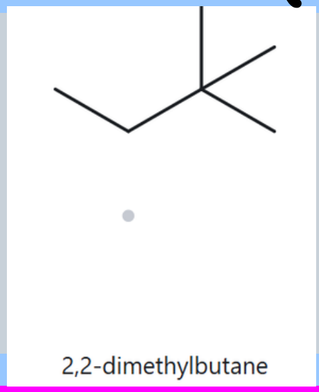
✓parent length matches ✓substituent positions match ✓formula agrees ✓back-check

Card B03 - Foundation: 2,2-dimethylbutane

Name Decoded

Parent: Butane → a 4-carbon chain.
Substituents: Two methyl groups, both on C2.
Expected Formula: 4+1+1=6 C; open-chain alkane → C_nH_{2n+2} → **C_6H_{14}**

Bond-line Structure (The Alkane)



Formula Check

Counted from the drawing: C_6H_{14}
Acyclic alkane: C_nH_{2n+2} = C_6H_{14} ✓

Back-Check

Returned by The Alkane: 2,2-dimethylbutane
✓ Same as the given name.

Verification Strip

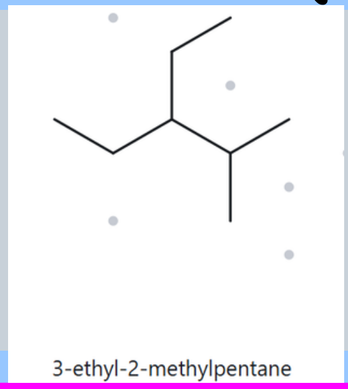
✓parent length matches ✓substituent positions match ✓formula agrees ✓back-check

Card B04 - Developing: 3-ethyl-2-methylpentane

Name Decoded

Parent: Pentane → a 5-carbon chain.
Substituents: Ethyl on C3; methyl on C2.
Expected Formula: 5+2+1=8 C; open-chain alkane → C_nH_{2n+2} → **C_8H_{18}**

Bond-line Structure (The Alkane)



Formula Check

Counted from the drawing: C_8H_{18}
Acyclic alkane: C_nH_{2n+2} = C_8H_{18} ✓

Back-Check

Returned by The Alkane: 3-ethyl-2-methylpentane
✓ Same as the given name.

Verification Strip

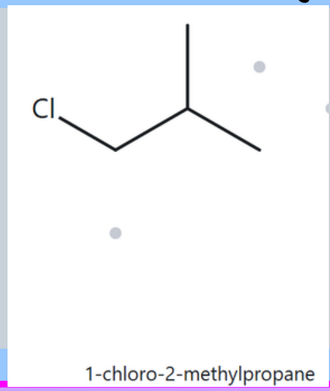
✓parent length matches ✓substituent positions match ✓formula agrees ✓back-check

Card B05 - Developing: 1-chloro-2-methylpropane

Name Decoded

Parent: Propane → a 3-carbon chain.
Substituents: Chloro on C1; methyl on C2.
Expected Formula: 3+1=4 C; chloro replaces one H on an acyclic alkane(C_4H_{10}) → **C_4H_9Cl**

Bond-line Structure (The Alkane)



1-chloro-2-methylpropane

Formula Check

Counted from the drawing: C_4H_9Cl
Acyclic haloalkane: $C_nH_{2n+1}Cl$ = C_4H_9Cl ✓

Back-Check

Returned by The Alkane: 1-chloro-2-methylpropane
✓ Same as the given name.

Verification Strip

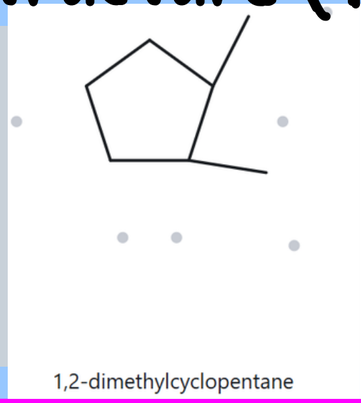
✓parent length matches ✓substituent positions match ✓formula agrees ✓back-check

Card B06 - Developing: 1,2-dimethylcyclopentane

Name Decoded

Parent: Cyclopentane → a 5-carbon ring.
Substituents: Methyl groups on C1 and C2.
Expected Formula: 5+2=7 C; single ring alkane → C_nH_{2n} → **C_7H_{14}**

Bond-line Structure (The Alkane)



Formula Check

Counted from the drawing: C_7H_{14}
Monocyclic alkane: C_nH_{2n} = C_7H_{14} ✓

Back-Check

Returned by The Alkane:
1,2-dimethylcyclopentane
✓ Same as the given name.

Verification Strip

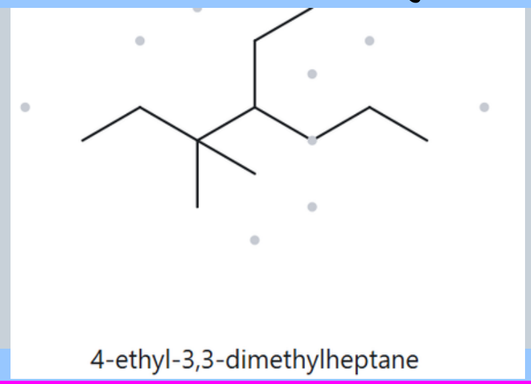
✓parent length matches ✓substituent positions match ✓formula agrees ✓back-check

Card B07 - Developing: 4-ethyl-3,3-dimethylheptane

Name Decoded

Parent: Heptane → a 7-carbon chain.
Substituents: Ethyl on C4; two methyl groups on C3.
Expected Formula: 7+2+1+1=11 C; open-chain alkane → C_nH_{2n+2} → **$C_{11}H_{24}$**

Bond-line Structure (The Alkane)



Formula Check

Counted from the drawing: $C_{11}H_{24}$
Acyclic alkane: C_nH_{2n+2} = $C_{11}H_{24}$ ✓

Back-Check

Returned by The Alkane:
4-ethyl-3,3-dimethylheptane
✓ Same as the given name.

Verification Strip

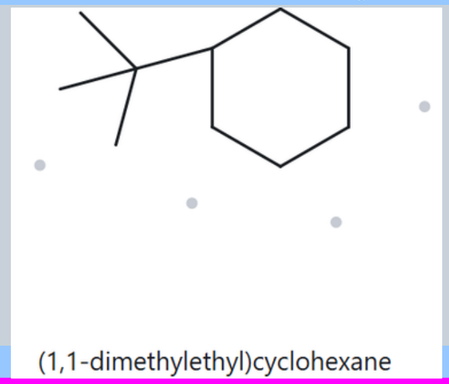
✓parent length matches ✓substituent positions match ✓formula agrees ✓back-check

Card B08 - Challenge: tert-butylcyclohexane

Name Decoded

Parent: Cyclohexane → a 6-carbon ring.
Substituents: tert-butyl on C1 (systematically: (1,1-dimethylethyl)).
Expected Formula: 6+4=10 C; single ring alkane → C_nH_{2n} → **$C_{10}H_{20}$**

Bond-line Structure (The Alkane)



Formula Check

Counted from the drawing: $C_{10}H_{20}$
Monocyclic alkane: C_nH_{2n} = $C_{10}H_{20}$ ✓

Back-Check

Returned by The Alkane:
(1,1-dimethylethyl)cyclohexane
✓ Same as the given name.
Note: The tool outputs the systematic bracketed form (1,1-dimethylethyl) for the common tert-butyl group, which is an accepted equivalent form earning full credit.

Verification Strip

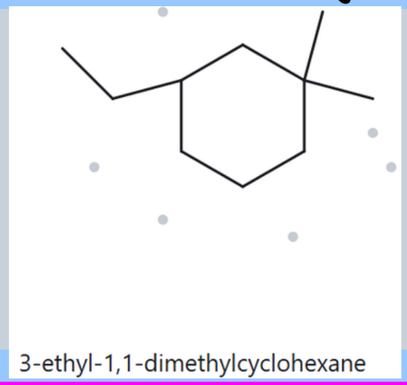
✓parent length matches ✓substituent positions match ✓formula agrees ✓back-check

Card B09 - Challenge: 3-ethyl-1,1-dimethylcyclohexane

Name Decoded

Parent: Cyclohexane → a 6-carbon ring.
Substituents: Ethyl on C3; two methyl groups on C1.
Expected Formula: 6+2+1+1=10 C; single ring alkane → C_nH_{2n} → **$C_{10}H_{20}$**

Bond-line Structure (The Alkane)



Formula Check

Counted from the drawing: $C_{10}H_{20}$
Monocyclic alkane: C_nH_{2n} = $C_{10}H_{20}$ ✓

Back-Check

Returned by The Alkane:
3-ethyl-1,1-dimethylcyclohexane
✓ Same as the given name.

Verification Strip

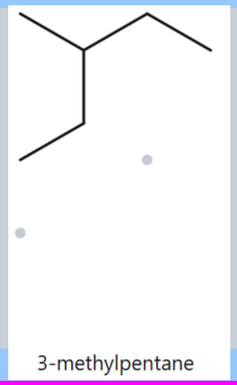
✓parent length matches ✓substituent positions match ✓formula agrees ✓back-check

Card B10 - Challenge: 2-ethylbutane

Name Decoded

Parent: Butane → a 4-carbon chain.
Substituents: Ethyl on C2
Expected Formula: 4+2=6 C; open-chain alkane → C_nH_{2n+2} → **C_6H_{14}**

Bond-line Structure (The Alkane)



Formula Check

Counted from the drawing: C_6H_{14}
Acyclic alkane: C_nH_{2n+2} = C_6H_{14} ✓

Back-Check

Returned by The Alkane:
3-methylpentane
✘ Not the same as the given name.
Reason: The given name "2-ethylbutane" violates the longest continuous chain rule. Tracing through the ethyl group creates a 5-carbon chain (pentane) with a single methyl branch at carbon-3.

Verification Strip

✓parent length matches ✓substituent positions match ✓formula agrees ✓back-check